

System Requirements Specification

Mk1 mini autonomous boat: modular GPS catamaran with Mission Control

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Document control

Issue	Date	Change	By
A	28 September 2026	First issue, for review. Includes the Concept of Operations (section 3) and Concept of Use (section 4).	Claude (drafted)

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: please check that every requirement is right (what you want), complete (nothing missing), testable, and has the right priority. Open items are listed in section 8.

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1. Introduction

1.1 Purpose

This specification defines **what** the Boaty Mk1 system must do and how each requirement will be verified. It is the input to architecture design and the reference for testing. It deliberately says as little as possible about how. Design choices from the concept report appear only where they are real constraints.

1.2 Scope

In scope: the Mk1 boat (hulls, thruster pods, mast, electronics, helm, mission computer, camera), the Mission Control bank station (Raspberry Pi 5), the mission-planning and validation software, the simulation environment, and the operating procedures for use at Milton Country Park.

Out of scope: salt water, rivers with current, night operation, operation beyond the operator's sight, carrying anything other than DUPLO bricks and the camera, and the design of a pure-Python helm (only its requirements, if Route B is chosen).

1.3 Conventions

Term / code	Meaning
shall	Mandatory. Priority M (Must): Mk1 is not complete without it.
should	Expected. Priority S (Should): include unless there's a good, recorded reason not to.
may	Optional. Priority C (Could): a nice-to-have or later upgrade.
ID	AREA- <i>nnn</i> , e.g. FS-004. IDs are never reused. Deleted requirements are marked 'deleted'.
Verification	T Test, D Demonstration, I Inspection, A Analysis, S Simulation
Stage	First stage the requirement is verified at: SIM, BENCH, POOL (paddling pool or small water), LAKE
Trace	The stakeholder need(s) the requirement serves (section 2.2)

1.4 References

Ref	Document
[1]	Boaty Concept Selection Report v1.1, 28 Sep 2026 (docs/concept)
[2]	ArduPilot Rover documentation (geofence, failsafes, SITL): ardupilot.org/rover
[3]	BS EN 71-1, Safety of toys: mechanical and physical properties (small-parts cylinder)
[4]	Ofcom IR 2030, UK interface requirements for licence-exempt short-range devices
[5]	Wildlife and Countryside Act 1981 (protection of wild birds and their nests)
[6]	Cambridge Sport Lakes Trust: Milton Country Park rules and conditions (to be obtained, TBD-03)

Terms and abbreviations are in Appendix A.

2. System overview

2.1 System context

Boaty is two physical units: the **boat** and the **Mission Control** bank station. They are linked by Wi-Fi. The adult operator and the 4-year-old crew use Mission Control only. It reaches the Claude API through a phone hotspot to turn spoken or typed instructions into missions.

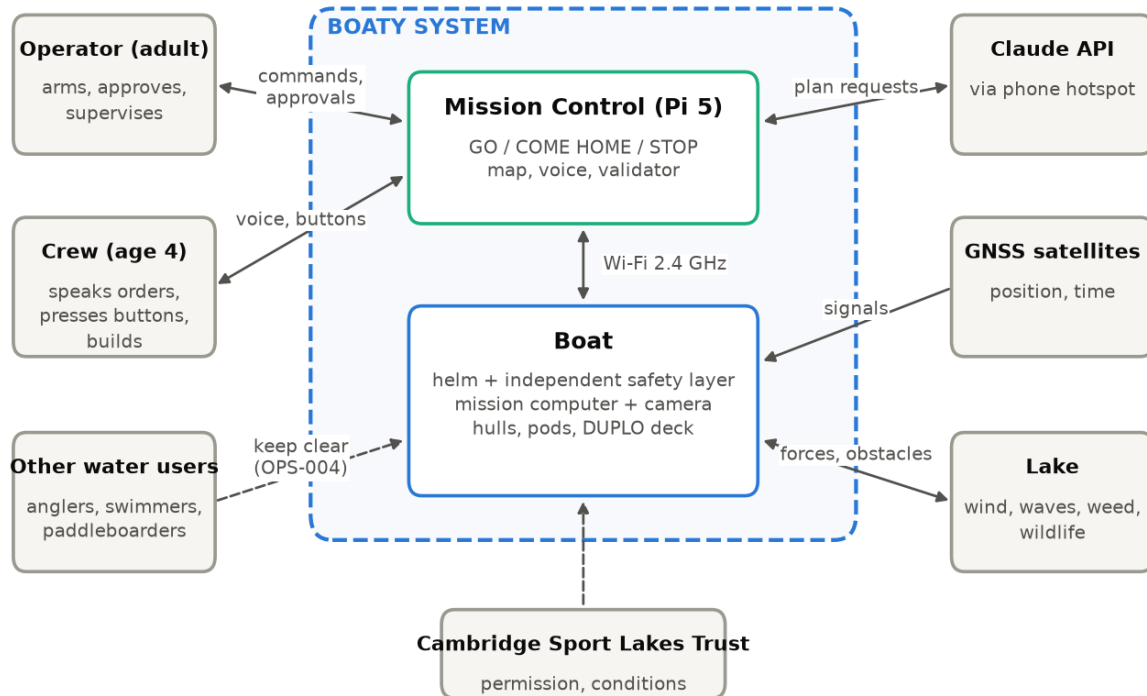


Figure 1. System context.

2.2 Stakeholder needs

ID	Need	Description
STK-01	Never lose the boat	The boat returns, or stays findable and reachable from our bank, whatever fails.
STK-02	Plain-English instructions	The crew can say what they want ("explore the bay, photograph some ducks, then come back") and the boat does it.
STK-03	Photos of ducks	The boat comes back with good photos of ducks and a record of the voyage.
STK-04	A build a 4-year-old can join	Modular, chunky, colourful parts he can assemble and customise, including building on it with DUPLO.
STK-05	Affordable	Around £100 cash for Mk1, reusing the Raspberry Pi 5 and Ultimaker printer.
STK-06	Safe for people and wildlife	No harm to the child, other water users or wildlife.
STK-07	Welcome at the lake	Operation is acceptable to Cambridge Sport Lakes Trust and other users of Milton Country Park.
STK-08	Python-first and extensible	Software the parent can read, change and extend in Python, including a future pure-Python helm.

2.3 Operating modes

Mode	Behaviour	Leaves by
DISARMED	Motors inhibited. Power-on state. Safe to handle.	Adult arms (MOD-003) with pre-arm checks passed.
HOLD	Armed. Keeps station near the hold point. After a STOP or a position/stuck failsafe the motors are off and the boat drifts.	GO → AUTO; COME HOME → RTL; adult → MANUAL or DISARMED.
MANUAL	Adult drives from Mission Control. Fence still enforced.	Adult → HOLD, AUTO or RTL; failsafes → HOLD or RTL.
AUTO	Executes the loaded, validated mission.	Mission end → RTL; failsafes → RTL or HOLD; STOP → HOLD (motors off).
RTL	Returns home inside the fence, avoiding exclusions.	Arrival → HOLD; failsafes → HOLD; STOP → HOLD (motors off).

Table 1. Boat modes (MOD-001).

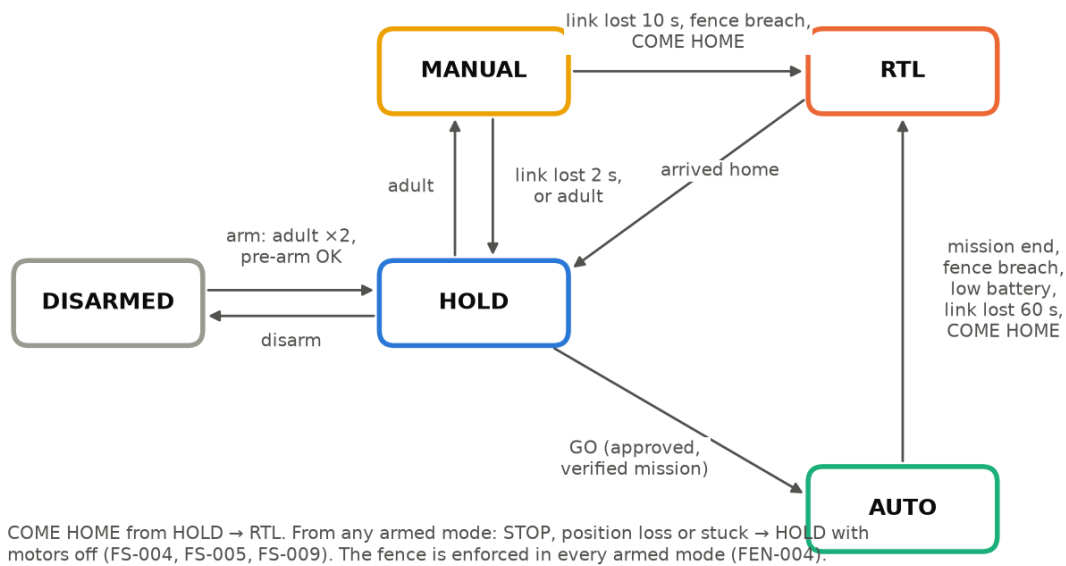


Figure 2. Mode transitions.

2.4 Design constraints carried from the concept

- Twin-hull catamaran, modular, 3D-printed on the owned Ultimaker; DUPLO-compatible deck.
- Two brains: fence and failsafes run on an independent processor (SAF-001). Everything the project writes is Python (SWE-001).
- **Software route is still open (TBD-01).** Route A: ArduPilot helm, with Python above it. Route B: pure-Python helm plus a MicroPython guardian. The requirements are written to hold for either route. Route-specific requirements are marked (SAF-005, MC-014).

2.5 Assumptions and dependencies

- The Claude API and mobile data are available on the bank. When they are not, the offline templates are used (MIS-005).
- GNSS is available in the open at the lake, with no tree canopy over the operating area.
- Cambridge Sport Lakes Trust grants permission, possibly with conditions that add requirements (TBD-03).
- Prices and part availability as in the concept report, $\pm 20\%$.

3. Concept of Operations (ConOps)

The ConOps describes the system from an operational point of view: why it is operated, where, by whom, through which phases, and how normal and abnormal situations are handled. Section 4 then describes the same system from the users' point of view.

3.1 Operational goal

Send a small, slow, unsinkable boat to explore a fenced area of a lake on a plain-English instruction, photograph the wildlife from a respectful distance, and bring it back to the bank every time. Do it as a shared parent-and-child activity that is safe, fun and welcome at the lake.

3.2 Operating sites and environment

Site	Use	Notes
Home (desk)	Simulation, planning, software, maintenance, charging	SITL and simulated camera (SWE-004)
Bath / paddling pool	Float, trim, driving, stuck and failsafe demonstrations	POOL stage. Child's main hands-on stage
Small private water (if available)	First autonomous missions within casting range	Optional, before Milton
Milton Country Park lake (Todd's Pit or Dickerson's Pit, TBD-02)	Real missions in a fenced 'home bay'	Shared water: anglers, swimmers, watersports. Needs permission (OPS-002)

Parameter	Operating envelope	Req.
Water	Fresh, still (lake or pond), waves ≤ 100 mm	ENV-003
Wind	≤ 5.4 m/s (Beaufort 3)	ENV-002
Air / water temperature	0-35 °C / 2-25 °C	ENV-001
Area	Inside the site fence. At Milton, Mk1 within 100 m of launch	OPS-006
Mission	≤ 20 min, $\leq 50\%$ of usable energy	MIS-003
Speed	Cruise 1.0 m/s, never above 1.5 m/s	NAV-002/003
Light	Daylight, boat in the operator's sight	1.2
People	No swim sessions or courses; ≥ 20 m from other water users	OPS-003/004

Table 2. Operating envelope.

3.3 Roles and responsibilities

Role	Who	Responsibilities
Operator	Parent	Accountable for safety. Gets permission. Runs the checklist. Arms, approves missions, supervises the crew and the boat, commands recovery, and never lets the child near the water unsupervised.
Crew	Son, age 4	Builds and customises the boat, gives the orders, presses GO / COME HOME / STOP, spots the boat, reviews photos.
Maintainer	Parent	Software, configuration, charging, repairs, spares, logs and FMEA upkeep.
Site owner	Cambridge Sport Lakes Trust	Grants permission and sets conditions.
Other water users	Anglers, swimmers, paddlers	Not system users. The system and the operator keep clear of them.

3.4 Operational phases

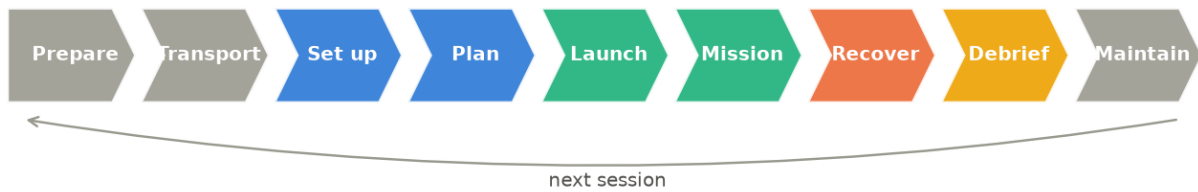


Figure 3. Operational phases of a session.

Phase	Activities	Boat mode	Key reqs
Prepare (home)	Charge the battery. Pull the tagged release. Load the site fence. Rehearse the plan in simulation. Pack the recovery and spares kits.	DISARMED	SWE-006, OPS-010/011
Transport	Boat in its carry cradle, battery separate.	Off	MEC-014, PWR-007
Set up (bank)	Check notices and other users. Power up Mission Control and the boat. Crew fits the pods and DUPLO crew. Complete the pre-launch checklist.	DISARMED	OPS-001/003, PRE-006
Plan	Crew gives the order. Mission generated, validated, shown and approved, then uploaded and verified.	DISARMED	NLI-*, VAL-*
Launch	Place the boat on the water. Adult arms (two-step). The boat holds.	HOLD	MOD-003/004, PRE-*
Mission	Crew holds GO. Boat runs the mission and takes photos. Operator monitors. COME HOME and STOP always available.	AUTO → RTL	MIS-*, FS-*, CAM-*
Recover	Boat holds at home. Adult disarms and lifts it out.	HOLD → DISARMED	MOD-005
Debrief	Photos download. Captain's log made. Crew finds the ducks.	DISARMED	CAM-007, DET-001/002
Maintain (home)	Rinse and dry. Battery to storage. Archive logs. Record issues. Update the FMEA if anything changed.	Off	OPS-010, LOG-*, SAF-006

3.5 Operational scenarios

One normal scenario and the contingencies the system must handle. Each contingency is rehearsed in simulation before the first lake trial (OPS-012). The end state is always 'boat at home' or 'boat stopped inside the fence and recoverable'.

#	Scenario / trigger	System response	Operator / crew action	Reqs
OS-1	Duck patrol (normal). Crew asks to explore and photograph ducks.	Plans, validates, survey with photo points, RTL, HOLD.	Approve; crew holds GO; watch; disarm; debrief.	MIS-001, VAL-*
OS-2	Manual drive (pool or near bank).	Adult drives; fence and speed limits still apply.	Adult drives; crew spots.	MC-012, FEN-004
OS-3	Link lost mid-mission.	Continues the mission; RTL if still lost after 60 s.	Watch; walk to a better spot; wait at home point.	FS-003, MC-009
OS-4	Low battery.	RTL at 35%; slow and alarm at 15%.	Watch it come home; end the session.	FS-001
OS-5	Stuck in weed.	HOLD, alarm, up to 3 reverse bursts, then resume or stay.	If still stuck: COME HOME to retry, else recover by line.	FS-005/006, OPS-008
OS-6	Position loss (GNSS).	Motors stop; RTL once healthy for 10 s.	Watch; recover by line if it doesn't recover.	FS-004
OS-7	Fence breach (wind, error).	RTL within 1 s; motors stop if far outside.	Watch; review logs at home.	FEN-005/006

#	Scenario / trigger	System response	Operator / crew action	Reqs
OS-8	Person nearby: angler, swimmer or paddler within 20 m.	No automatic detection.	Operator presses COME HOME or STOP.	OPS-004, FS-009
OS-9	Dead in the water (total power loss).	Passive: floats, flag and hi-vis visible.	Wait for drift to the bank; pole and net, or cast a line over the hoop.	REC-001..005, OPS-008
OS-10	No internet.	Natural-language planning unavailable; offline templates offered.	Pick a template with the crew.	MIS-005, MC-011
OS-11	Mission computer crash.	Helm completes the mission and RTL; no photos.	Reboot after recovery; check logs.	FS-007, SAF-002

3.6 Support concept

- **Maintenance:** rinse and dry the motors after every session. Check props, guards and seals. Motors and props are cheap consumables. Keep spares at home and a field kit in the bag (OPS-011).
- **Software:** changes are made and tested in simulation at home, and a tagged release is taken to the lake. No field edits to safety parameters (SAF-007, SWE-006).
- **Training:** the operator learns and rehearses in simulation (OPS-012). The crew learns by playing with Mission Control in simulation mode at home.
- **Records:** logs, photos and captain's logs are archived per session. Incidents feed the FMEA.

3.7 Evolution

Mk1 grows by earned steps: a bigger fence after 5 good missions (OPS-006); on-board duck spotting (DET-003); an independent radio (COM-007); and, if chosen, the pure-Python helm behind the same interface (SWE-003).

4. Concept of Use (ConUse)

The ConUse describes how each user interacts with the system: who they are, the principles that shape the interaction, what they can touch and say, and the use cases they perform.

4.1 User profiles

User	Profile	Needs from the interaction
Operator	Adult. Mechanical engineer, confident with electronics and Python. Attention is split between a 4-year-old, the bank and the boat.	Glanceable status, loud alarms, few steps, nothing hidden. Firm control of anything hazardous. Full logs afterwards.
Crew	Age 4. Can't read yet. Short attention (about 20 min). Loves building, big buttons and ducks. Needs to feel it's his boat.	Big colourful controls, immediate sound and light feedback, simple spoken language, and a guaranteed happy ending: photos.
Onlookers	Curious park visitors, other children.	Nothing to operate. The boat is clearly supervised, slow, and labelled (REC-006).

4.2 Interaction principles

- **Safe whatever the crew presses.** COME HOME and STOP can never cause harm. GO works only when an adult has made it safe (MC-003, VAL-008).
- **No reading needed.** Colour, icon, position and voice carry every crew interaction (CHD-005).
- **Adult gate on hazards.** Arming, fences, approvals and parameters need the adult key or PIN (MC-008).
- **The system talks.** Every mode change and event is spoken in child-friendly words (MC-007).
- **Always end on a win.** Short missions that come home with photos, and a captain's log to look at (MIS-003, DET-002).

4.3 Mission Control: what the crew can touch and say

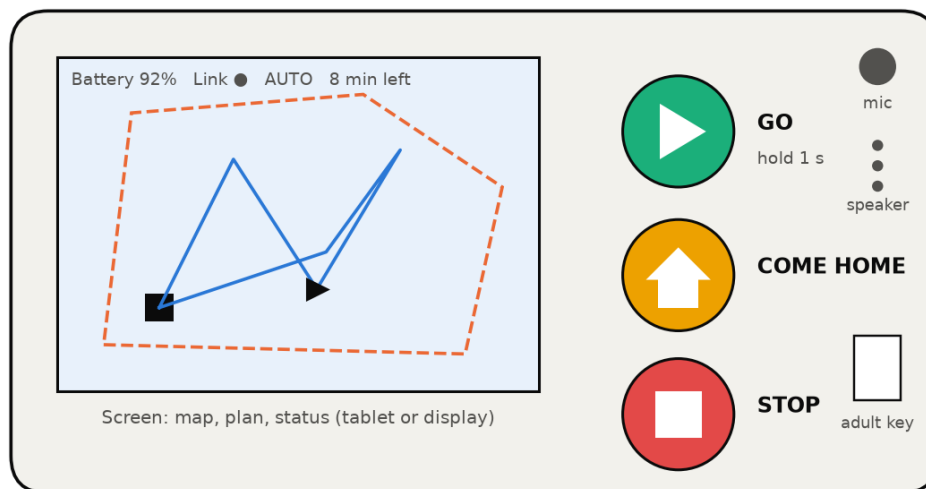


Figure 4. Mission Control panel concept (layout indicative).

Control	What it does	Available when	Feedback	Why it's safe
GO (green, ►)	Starts the approved mission	Armed in HOLD with an approved, verified mission	Light on; "Off we go!"	Adult pre-approved; hold 1 s
COME HOME (yellow, house)	RTL	Any armed mode	"Coming home!"	RTL stays inside the fence
STOP (red, ■)	Motors off, boat drifts	Any time	"Stopping!"	Removes energy
Voice (push-to-talk)	Asks for a mission plan	DISARMED or HOLD	Plan shown on the map and spoken	Validator plus adult approval

Control	What it does	Available when	Feedback	Why it's safe
DUPLO deck	Crew, cargo, lookout	Boat disarmed	It's his	Away from props and electronics
Adult key / PIN	Arm, approve, fence, settings	Operator only	Screen and voice	Gates every hazard

4.4 Use cases

UC	Use case	Actor	Main flow	Reqs
UC-01	Plan a mission by voice	Crew + operator	Crew speaks, then the plan appears and is spoken. Operator approves. Upload and read-back.	NLI-*, VAL-*
UC-02	Start a mission	Crew	Adult arms, then the crew holds GO. The boat departs and the system announces it.	MC-003/004, MOD-006
UC-03	Call the boat home	Crew or operator	Press COME HOME. RTL, then HOLD at home.	MC-003, NAV-007
UC-04	Emergency stop	Anyone at the panel	Press STOP. Motors stop within 1 s.	FS-009
UC-05	Drive manually	Operator	Select MANUAL and drive with the joystick or gamepad; fence enforced.	MC-012, FEN-004
UC-06	Set up the site fence	Operator	Draw the inclusion fence and exclusions on the map, then save and commit.	FEN-001..003
UC-07	Pre-launch checklist	Operator (crew helps)	Step through the items and sign off. Arming is then allowed.	OPS-001, PRE-006
UC-08	Review photos and captain's log	Crew + operator	Photos download. Ducks highlighted. Captain's log read aloud.	CAM-007, DET-002, MC-015
UC-09	Build and customise	Crew (operator helps)	Clip hull segments and pods together, then build on the DUPLO deck.	MEC-007/009, CHD-*
UC-10	Practise in simulation	Operator + crew	Mission Control in sim mode with the same buttons and voice.	SWE-004, OPS-012
UC-11	Charge and maintain	Operator	Battery out, charged in the fire-safe bag, rinse and dry, archive logs.	PWR-007, OPS-010

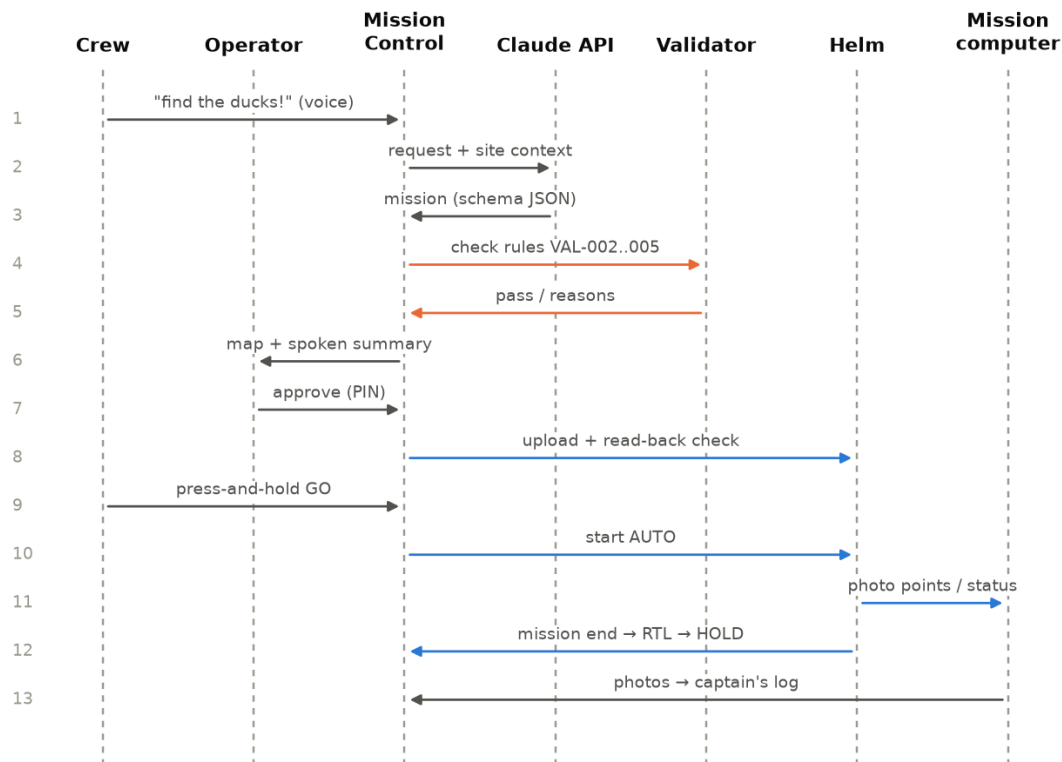


Figure 5. UC-01 and UC-02 in detail: from spoken instruction to captain's log. The validator (orange) and the helm (blue) are the safety gates.

4.5 A day at the lake: a use story

Saturday morning at Milton. The notice board shows no algae warning and no swim session, and the wind is light and blowing towards the bank. Dad works through the checklist on the screen while his son clips the orange pods on with the big thumb-screws and sits two DUPLO ducks on the deck as 'lookouts'. He presses the talk button: "Go and find the real ducks and take pictures!" A few seconds later a blue path appears on the map and the box says, "I'll explore the bay for ten minutes, take twenty pictures, then come home." Dad checks the route, turns the key and approves it. The boat goes in the water, Dad arms it, and his son holds the green button until it says "Off we go!"

Halfway round, the boat stops and says "I'm stuck in some weed, trying to wiggle free." It reverses twice and carries on. An angler sets up nearby, so Dad lets his son press the yellow house: "Coming home!" The boat returns and waits by the bank. Back at the car they flick through the photos on the screen. Three have ducks, and the captain's log tells the story of the voyage.

5. Requirements

184 requirements in 22 areas. Columns: priority (Pri), verification method (Ver), first verification stage (Stage), and the stakeholder needs served (Trace, STK- prefix omitted).

5.1 Operating modes (MOD)

The boat's behaviour is defined by a small set of modes (see section 2.3). These are the only states the system may be in.

ID	Requirement	Pri	Ver	Stage	Trace
MOD-001	The system shall implement exactly these boat modes: DISARMED, HOLD, MANUAL, AUTO and RTL, as defined in section 2.3.	M	I, S	SIM	01, 06
MOD-002	On power-up the boat shall enter DISARMED, with both motors inhibited.	M	T	BENCH	06
MOD-003	Arming shall require two deliberate adult actions: the arming switch on the boat set to ARM, and arming confirmed at Mission Control.	M	D	BENCH	06
MOD-004	The system shall refuse to arm unless every pre-arm check (PRE-001 to PRE-008) passes, and shall state the failing check in plain language at Mission Control.	M	T	SIM	01, 06
MOD-005	When an AUTO mission completes, the boat shall enter RTL. On reaching home (within the waypoint acceptance radius) it shall enter HOLD.	M	S, T	SIM	01
MOD-006	While armed, the operator shall be able to command HOLD, RTL, MANUAL or STOP from Mission Control at any time, overriding AUTO.	M	T	SIM	01, 06
MOD-007	Every mode change shall be shown and announced at Mission Control within 2 s, whatever caused it.	S	T	SIM	01, 04
MOD-008	The boat shall disarm automatically after 60 s in HOLD at home with no operator input, and announce it.	C	S	SIM	06

5.2 Pre-arm checks (PRE)

Checks the helm and Mission Control perform before arming is allowed (MOD-004).

ID	Requirement	Pri	Ver	Stage	Trace
PRE-001	GNSS shall have a 3D fix with at least 8 satellites and HDOP no greater than 1.5.	M	T	BENCH	01
PRE-002	A geofence shall be loaded and enabled, and home shall lie inside the inclusion fence and outside all exclusion zones.	M	T	SIM	01, 07
PRE-003	Battery state of charge shall be at least 80%.	M	T	BENCH	01
PRE-004	The heading sensor shall report healthy and calibrated.	M	T	BENCH	01
PRE-005	For AUTO: a mission that has passed validation and operator approval shall be loaded, and read back from the helm to match (VAL-010).	M	T	SIM	01, 02
PRE-006	The pre-launch checklist (OPS-001) shall be complete and signed off at Mission Control for this session.	M	D	POOL	06, 07
PRE-007	The independent safety layer (SAF-001) shall report healthy.	M	T	BENCH	01
PRE-008	Logging shall be active on the helm and Mission Control, each with at least 1 GB free.	M	T	BENCH	01

5.3 Navigation and control (NAV)

ID	Requirement	Pri	Ver	Stage	Trace
NAV-001	The boat shall steer by differential thrust from two thrusters, and be able to go forward and in reverse.	M	D	POOL	01
NAV-002	Cruise speed in calm water shall be 1.0 m/s $\pm$ 0.2 m/s.	M	T	LAKE	03, 06
NAV-003	Speed through the water shall be limited to 1.5 m/s in every mode.	M	T	POOL	06, 07

ID	Requirement	Pri	Ver	Stage	Trace
NAV-004	In AUTO the boat shall pass within the acceptance radius of each waypoint. The radius is configurable from 1 to 10 m, default 3 m.	M	S, T	SIM	02
NAV-005	Cross-track error between waypoints shall be no more than 3 m RMS in wind up to 3 m/s.	S	T	LAKE	01, 02
NAV-006	In HOLD with station-keeping, the boat shall stay within 5 m of the hold point in wind up to 3 m/s.	S	T	LAKE	01, 03
NAV-007	In RTL the boat shall navigate to home without entering any exclusion zone or leaving the inclusion fence.	M	S, T	SIM	01, 06
NAV-008	Above 0.3 m/s the heading estimate shall use GNSS course as well as the magnetometer, so a disturbed compass cannot alone cause a wrong course.	S	S, T	SIM	01

5.4 Geofence (FEN)

The geofence keeps the boat inside a known safe area of water. It is enforced by the independent safety layer (SAF-001).

ID	Requirement	Pri	Ver	Stage	Trace
FEN-001	The system shall support one inclusion polygon of 3 to 70 vertices and up to 10 exclusion zones, each a polygon or circle.	M	S	SIM	01, 06
FEN-002	Fences shall be drawn on a map at Mission Control, stored per site with a name, and version-controlled.	M	D	SIM	01, 07
FEN-003	Mission Control shall show a 5 m guide inside the drawn inclusion fence, and warn if any home or waypoint falls within it.	S	D	SIM	01
FEN-004	The fence shall be enforced in every armed mode, including MANUAL.	M	S, T	SIM	01, 06
FEN-005	On a fence breach the boat shall enter RTL within 1 s of the breach.	M	S, T	SIM	01
FEN-006	If the boat stays outside the inclusion fence for more than 30 s, or gets more than 10 m outside it, the motors shall stop (HOLD, no thrust) and an alarm shall sound at Mission Control.	M	S	SIM	01, 06
FEN-007	Fence and exclusion zones shall not be changed while the boat is armed.	M	T	SIM	01

5.5 Failsafes (FS)

Automatic responses to faults, summarised in the table at the end of this section. Thresholds are initial values, to be tuned in trials (TBD-08).

ID	Requirement	Pri	Ver	Stage	Trace
FS-001	Low battery: when estimated state of charge falls to 35%, the boat shall enter RTL. At 15% it shall continue RTL at reduced speed and sound an alarm.	M	S, T	SIM	01
FS-002	Link loss in MANUAL: with no command link for 2 s the boat shall enter HOLD, and after 10 s it shall enter RTL.	M	S, T	SIM	01
FS-003	Link loss in AUTO: the boat shall continue the mission. If the link is still lost after 60 s it shall enter RTL.	M	S, T	SIM	01
FS-004	Position loss: if the GNSS fix is lost, HDOP exceeds 2.5 or the position estimate is unhealthy for 3 s, the motors shall stop (HOLD, no thrust). After 10 s of healthy position the boat shall enter RTL.	M	S	SIM	01
FS-005	Stuck detection: if commanded thrust is at least 50% and ground speed stays below 0.1 m/s for 5 s, the boat shall enter HOLD and alarm.	M	S, T	POOL	01
FS-006	Weed-shedding: after a stuck detection the boat shall try up to 3 short reverse bursts, then resume its previous mode if free, or stay in HOLD if not.	S	T	POOL	01

ID	Requirement	Pri	Ver	Stage	Trace
FS-007	Mission computer failure: if the helm receives no mission computer heartbeat for 3 s, it shall complete the loaded mission and RTL on its own.	M	S, T	SIM	01
FS-008	Helm failure: if motor commands stop arriving from the helm, the motors shall stop within 1 s.	M	T	BENCH	01, 06
FS-009	STOP: pressing STOP at Mission Control shall stop both motors within 1 s, whatever the mode.	M	T	BENCH	06
FS-010	Water ingress: a moisture sensor in the electronics box shall trigger RTL and an alarm.	S	T	BENCH	01
FS-011	When several failsafes are active, the most conservative response shall win: motors stopped over RTL over continuing the mission.	M	S	SIM	01, 06
FS-012	Every failsafe event shall be logged, and announced at Mission Control within 2 s if the link is up.	M	S, T	SIM	01
FS-013	No single failure shall cause the boat to leave the inclusion fence under its own power. This is shown by FMEA (SAF-007) and by simulated fault injection.	M	A, S	SIM	01, 06

Condition	Detection	Response	Req.
Fence breach	Position outside inclusion / inside exclusion	RTL within 1 s	FEN-005
Far outside fence	> 30 s outside, or > 10 m outside	Motors stop, alarm	FEN-006
Low battery	SoC $\leq$ 35%	RTL	FS-001
Critical battery	SoC $\leq$ 15%	Continue RTL slowly, alarm	FS-001
Link loss (MANUAL)	No link 2 s / 10 s	HOLD, then RTL	FS-002
Link loss (AUTO)	No link 60 s	Continue, then RTL	FS-003
Position loss	No fix, HDOP > 2.5 or unhealthy for 3 s	Motors stop; RTL after 10 s healthy	FS-004
Stuck / weed	Thrust $\geq$ 50%, speed < 0.1 m/s for 5 s	HOLD, alarm, up to 3 reverse bursts	FS-005/006
Mission computer down	No heartbeat 3 s	Helm completes mission and RTL	FS-007
Helm down	Motor commands stop	Motors stop $\leq$ 1 s	FS-008
STOP pressed	Operator	Motors stop $\leq$ 1 s	FS-009
Water in box	Moisture sensor	RTL, alarm	FS-010

Table 3. Failsafe summary (initial thresholds, TBD-08).

5.6 Safety architecture (SAF)

How the safety functions are protected from the rest of the system. These requirements hold for both software routes (section 2.4).

ID	Requirement	Pri	Ver	Stage	Trace
SAF-001	Fence and failsafe functions shall run on a processor separate from the mission computer and Mission Control. That processor shall run bare-metal firmware or an RTOS, not a general-purpose operating system.	M	I	BENCH	01, 08
SAF-002	Those functions shall keep working with the mission computer, Mission Control and the radio link all failed.	M	S, T	SIM	01
SAF-003	The mission computer and Mission Control shall command the helm only through the helm interface (SWE-002). They shall not be able to disable the fence or failsafes while armed.	M	I, T	SIM	01, 08
SAF-004	Output from the language model shall never be sent to the helm as a command. Only mission objects that have passed validation (VAL-001) may reach the helm.	M	I, T	SIM	01, 02

ID	Requirement	Pri	Ver	Stage	Trace
SAF-005	Route B only: a MicroPython guardian with its own GNSS shall pass motor commands through only while the helm heartbeat is fresh (under 0.5 s) and the boat is inside the fence. Otherwise it shall stop the motors or run its own return-home routine. Applies only if Route B is chosen (TBD-01).	M	T	BENCH	01, 08
SAF-006	A failure modes and effects analysis (FMEA) shall be done before the first lake trial and updated after any change to hardware or safety software.	M	A	SIM	01, 06
SAF-007	Safety parameters (fence, failsafe thresholds, speed limits) shall be version-controlled. The values loaded on the helm shall be checked against the committed baseline at pre-arm.	S	T	SIM	01

5.7 Missions (MIS)

A mission is a structured, versioned plan the helm can execute. Missions can come from the language model, templates or the map editor. All of them go through the same validator (section 5.9).

ID	Requirement	Pri	Ver	Stage	Trace
MIS-001	The system shall support these mission elements: waypoint, area survey (lawnmower pattern over a polygon, with lane spacing), photo point (hold, then take N photos), and return home.	M	S	SIM	02, 03
MIS-002	Every mission shall end with return home.	M	S	SIM	01
MIS-003	Planned mission duration shall not exceed 20 min (configurable up to 30 min). Planned energy shall not exceed 50% of usable battery energy.	M	A, S	SIM	01
MIS-004	Missions shall be represented in a documented JSON schema with a version number.	M	I	SIM	02, 08
MIS-005	Offline templates shall be available without internet, at least: 'Explore the bay', 'Duck patrol' (survey with photo points) and 'Lap of the bay'.	M	D	SIM	02, 04
MIS-006	An adult shall be able to create and edit missions on the map at Mission Control.	S	D	SIM	02

5.8 Natural-language instructions (NLI)

ID	Requirement	Pri	Ver	Stage	Trace
NLI-001	Mission Control shall accept instructions as typed text.	M	D	SIM	02
NLI-002	Mission Control shall accept spoken instructions through a microphone, using push-to-talk.	S	D	SIM	02, 04
NLI-003	Instructions shall be turned into a mission through the Claude API. The request shall include the site context: fence, exclusion zones, home, battery state, time and energy limits, and the mission schema.	M	T	SIM	02
NLI-004	The language model's reply shall be constrained to the mission schema (structured output or tool use). A malformed reply shall be rejected and retried at most twice. After that the user shall be offered the templates.	M	T	SIM	02, 01
NLI-005	The proposed mission shall be shown on the map with a one-sentence plain-language summary, which shall also be spoken aloud.	M	D	SIM	02, 04
NLI-006	Requests the boat must not or cannot do (for example leaving the fence, chasing animals, going to another lake) shall be declined in child-friendly words, with a safe alternative offered.	M	T	SIM	02, 06
NLI-007	With mobile data available, a plan shall be displayed within 20 s of the instruction.	S	T	LAKE	02
NLI-008	The API key shall be stored only on Mission Control and shall never be committed to the repository.	M	I	SIM	08

5.9 Mission validation (VAL)

A deterministic Python validator stands between every mission source and the helm. It is the main safeguard against a bad plan, wherever the plan came from.

ID	Requirement	Pri	Ver	Stage	Trace
VAL-001	Every mission, from any source, shall pass the validator before upload. There shall be no other path for uploading missions to the helm.	M	I, T	SIM	01, 02
VAL-002	Every waypoint, and every leg sampled at 1 m intervals or finer, shall be inside the inclusion fence by at least 3 m and outside every exclusion zone by at least 3 m.	M	T	SIM	01, 06
VAL-003	The mission shall meet the duration and energy limits in MIS-003.	M	T	SIM	01
VAL-004	The mission shall start within 10 m of home and end with return home.	M	T	SIM	01
VAL-005	Commanded speeds shall be within NAV-003. Each hold shall be 60 s or less. The photo count shall fit in the free camera storage. The waypoint count shall be within the helm's limit.	M	T	SIM	01, 03
VAL-006	The validator shall be pure Python with no network access and no randomness. It shall have a unit-test suite that includes adversarial missions, with at least 95% branch coverage.	M	T	SIM	01, 08
VAL-007	Each rejection shall be reported in plain language, naming the rule broken and where on the map.	M	D	SIM	02
VAL-008	After validation, an adult shall approve the mission at Mission Control. GO shall stay disabled until then.	M	T	SIM	01, 06
VAL-009	Any change to an approved mission shall void the approval.	M	T	SIM	01
VAL-010	After upload, the mission shall be read back from the helm and compared with the approved mission by checksum before GO is enabled.	M	T	SIM	01

5.10 Camera and photos (CAM)

ID	Requirement	Pri	Ver	Stage	Trace
CAM-001	The boat shall carry a forward-looking camera that captures still images of at least 2 MP (M) and at least 5 MP (S).	M	I, T	BENCH	03
CAM-002	Capture modes shall be: interval (every 2 to 30 s), at photo points, and on command.	M	T	BENCH	03
CAM-003	Every photo shall be tagged with UTC time, position, heading and mission ID.	M	T	BENCH	03
CAM-004	Local storage shall hold at least 500 photos. Full storage shall not affect navigation or safety.	M	T	BENCH	03, 01
CAM-005	A duck (about 0.3 m long) at 5 m shall span at least 40 pixels in the image.	S	A, T	POOL	03
CAM-006	The lens shall be shielded from spray and cleanable on the bank without tools.	S	I	POOL	03
CAM-007	After a mission, photos shall transfer to Mission Control automatically when in range, with 200 photos taking 5 min or less.	S	T	POOL	03
CAM-008	Mission Control may show a low-rate live view from the camera.	C	D	POOL	04

5.11 Duck finding and captain's log (DET)

ID	Requirement	Pri	Ver	Stage	Trace
DET-001	After a mission, the system shall identify photos containing waterfowl, with at least 80% precision on a test set of lake photos.	S	T	LAKE	03
DET-002	The system shall produce a captain's log: up to 6 best photos, a map of the route, distance, duration, ducks seen, and a short child-friendly story of the voyage.	S	D	LAKE	03, 04
DET-003	An upgraded mission computer may detect waterfowl on board in real time and trigger 'pause and look' (hold and take a burst of photos).	C	T	POOL	03

ID	Requirement	Pri	Ver	Stage	Trace
DET-004	If DET-003 is implemented, 'pause and look' shall hold position and shall never steer towards the detected animal. Conditional on DET-003.	M	S, T	SIM	06

5.12 Mission Control (bank station) (MC)

The Raspberry Pi 5 based ground station: the only interface the crew uses.

ID	Requirement	Pri	Ver	Stage	Trace
MC-001	Mission Control shall run on the owned Raspberry Pi 5 and operate for at least 2.5 h from a USB-C power bank.	M	T	BENCH	05
MC-002	It shall have three physical push buttons, each at least 30 mm across, in distinct colours with icons: GO (green), COME HOME (yellow), STOP (red).	M	I	BENCH	04, 06
MC-003	STOP and COME HOME shall always be active while armed. GO shall be active only when a validated, approved and verified mission is loaded and the boat is armed in HOLD.	M	T	SIM	01, 06
MC-004	GO shall require a 1 s press-and-hold, to prevent accidental starts.	S	T	BENCH	06
MC-005	The display shall show, updated at least once a second: a map with the boat's position, heading and track; fence, exclusions, home and mission; battery %, link status, mode and mission time remaining.	M	D	SIM	01, 02
MC-006	The display shall be readable in daylight. A tablet or phone used as the display is acceptable.	S	D	LAKE	01
MC-007	Mission Control shall give spoken announcements of mode changes, failsafes and mission events in child-friendly words.	S	D	SIM	04
MC-008	Adult-only functions (arming, fence editing, parameter changes, mission approval) shall be protected by a PIN or key switch.	M	T	SIM	06
MC-009	The last known position and track shall stay displayed after link loss, in a 'find my boat' view.	M	T	SIM	01
MC-010	Mission Control shall host the pre-launch checklist (OPS-001) and record who signed it off, and when.	M	D	SIM	06, 07
MC-011	Everything except natural-language planning and cloud photo analysis shall work without internet access.	M	T	SIM	01, 02
MC-012	An adult shall be able to drive the boat in MANUAL from Mission Control, using an on-screen joystick or a gamepad.	S	D	POOL	01
MC-013	The Mission Control case shall resist splashes (IPX4 intent).	S	I	BENCH	01
MC-015	Mission Control shall have a photo-review mode the crew can use: large thumbnails, stepped through with the buttons, and ducks highlighted.	S	D	SIM	03, 04
MC-014	Route A only: QGroundControl or MAVProxy shall be usable as a backup ground station.	S	D	SIM	01

5.13 Communications (COM)

ID	Requirement	Pri	Ver	Stage	Trace
COM-001	The boat-to-bank link shall use licence-exempt 2.4 GHz Wi-Fi within UK power limits.	M	I	BENCH	07
COM-002	Link range shall be at least 100 m line of sight over water, with the boat antenna at least 150 mm above the water.	M	T	LAKE	01
COM-003	Within range: telemetry at 1 Hz or faster, and command latency of 1 s or less (95th percentile).	M	T	POOL	01
COM-004	Link quality shall be shown at Mission Control and logged.	S	D	POOL	01
COM-005	Mission Control shall reach the internet through a phone hotspot. Losing internet shall not affect the boat link.	M	T	POOL	01, 02

ID	Requirement	Pri	Ver	Stage	Trace
COM-006	The link shall be secured with WPA2 or better and non-default credentials. The boat shall accept commands only from the authenticated Mission Control.	M	T	BENCH	01, 06
COM-007	An independent manual-override radio link (e.g. ExpressLRS) may be added.	C	D	POOL	01

5.14 Recoverability and locating (REC)

Passive measures that work with no power and no software.

ID	Requirement	Pri	Ver	Stage	Trace
REC-001	The boat shall stay afloat, carrying its maximum payload, with the electronics box flooded and any two hull segments breached.	M	T	POOL	01
REC-002	At least 50% of the visible top surface shall be fluorescent orange or yellow.	M	I	BENCH	01
REC-003	A flag at least 100 × 60 mm shall fly with its top at least 300 mm above the waterline.	M	I	BENCH	01
REC-004	An LED beacon shall be visible from 100 m in overcast daylight, with flash patterns showing armed, failsafe and low-battery states.	S	T	LAKE	01
REC-005	A recovery hoop of at least 60 mm internal diameter shall be fitted to the mast. It shall withstand a 50 N pull, and the boat shall tow by it without capsizing.	M	T	POOL	01
REC-006	The hull shall carry an 'If found' label with a contact phone number.	M	I	BENCH	01, 07
REC-007	A Bluetooth item tracker may be carried in the hull.	C	I	BENCH	01

5.15 Mechanical (MEC)

ID	Requirement	Pri	Ver	Stage	Trace
MEC-001	The boat shall be a twin-hull catamaran no more than 650 mm long, 400 mm wide and 500 mm high (including mast).	M	I	BENCH	04, 05
MEC-002	Ready-to-sail mass, excluding payload, shall be no more than 1.8 kg.	M	T	BENCH	01
MEC-003	The boat shall carry at least 300 g of payload on the deck with at least 50 mm of freeboard.	M	T	POOL	04
MEC-004	Static heel shall be 10° or less with 300 g at the deck edge. The boat shall not capsize in full-speed turns or in 100 mm waves.	M	T	POOL	01
MEC-005	Every printed part shall fit a 200 × 200 × 200 mm build volume (Ultimaker).	M	I	BENCH	05
MEC-006	Wetted and exposed printed parts shall be PETG or ASA. Hull segments shall be filled with closed-cell foam.	M	I	BENCH	01
MEC-007	Hull segments, thruster pods, mast, electronics box and DUPLO deck shall be removable and refittable by hand, without tools.	M	D	BENCH	04
MEC-008	Swapping a thruster pod shall take under 2 min. Pod connectors shall be waterproof and keyed.	S	D	BENCH	04
MEC-009	The deck plate shall carry LEGO DUPLO-compatible studs. Genuine bricks shall grip firmly and be removable by a 4-year-old.	M	T	BENCH	04
MEC-010	Propellers shall be guarded so that an 8 mm diameter probe cannot touch a blade.	M	T	BENCH	06
MEC-011	Thruster pods shall include weed-shedding features: a swept leading edge and a guard shape that sheds weed.	S	T	LAKE	01
MEC-012	The electronics box shall let in no water after 30 min at 0.3 m depth. Every cable entry shall go through a gland.	M	T	BENCH	01
MEC-013	The GNSS and compass shall be mounted at least 150 mm from motor and power wiring.	M	I	BENCH	01

ID	Requirement	Pri	Ver	Stage	Trace
MEC-014	The boat shall fit in a car boot and have a carry handle for one adult hand.	S	I	BENCH	05
MEC-015	Fasteners shall be stainless steel or plastic.	S	I	BENCH	01

5.16 Electrical and power (PWR)

ID	Requirement	Pri	Ver	Stage	Trace
PWR-001	Propulsion and avionics shall run from one removable 3S (nominal 11.1 V) lithium battery of 25 to 50 Wh.	M	I	BENCH	05
PWR-002	The battery shall be protected against over-discharge, over-current and short circuit (BMS or equivalent).	M	I, T	BENCH	06
PWR-003	A main fuse shall sit within 50 mm of the battery positive, rated no more than twice the maximum continuous current.	M	I	BENCH	06
PWR-004	A waterproof main power switch shall be operable without opening the electronics box.	M	I	BENCH	06
PWR-005	Avionics shall have a separate regulated 5 V supply. A full reverse-to-forward thrust step shall not reset or brown out the helm.	M	T	BENCH	01
PWR-006	Battery voltage and current shall be measured, logged and used for state of charge and failsafes.	M	T	BENCH	01
PWR-007	The battery shall be removed from the boat for charging, and charged only off-boat.	M	I	BENCH	06
PWR-008	All power connectors shall be polarised and keyed, with no two same-type connectors doing different jobs.	M	I	BENCH	06
PWR-009	Endurance at cruise speed from a full battery at 15 °C shall be at least 40 min (M), with a target of 60 min (S).	M	T	LAKE	01, 03
PWR-010	Electronics in the sealed box shall run for 60 min at 30 °C ambient in direct sun without fault or throttling.	M	T	BENCH	01
PWR-011	A power budget shall be kept up to date alongside the design.	S	A	BENCH	01

5.17 Software (SWE)

ID	Requirement	Pri	Ver	Stage	Trace
SWE-001	All project-written software shall be Python: CPython 3.11+ on Linux and MicroPython on microcontrollers. Third-party firmware (ArduPilot, ESC firmware) is excluded.	M	I	SIM	08
SWE-002	A Python helm interface shall provide: arm, disarm, set mode, upload mission, read mission back, start, hold, return home, stop, set fence, and a status stream.	M	I, T	SIM	08
SWE-003	Mission-level code shall run unchanged against any helm implementation, selected by configuration: first ArduPilot over MAVLink, later a Python helm.	M	T	SIM	08
SWE-004	The whole system shall run in simulation (helm simulator and simulated camera) on the Pi 5 or a laptop, with Mission Control software unchanged.	M	D	SIM	08, 01
SWE-005	Each failsafe requirement (FS-001 to FS-013) shall have an automated simulation scenario.	M	T	SIM	01
SWE-006	Source, configuration and safety parameters shall live in this git repository, with a tagged release for each lake trial.	M	I	SIM	08
SWE-007	Unit tests (pytest), linting (ruff) and type checking (mypy) shall run automatically on every push.	S	I	SIM	08
SWE-008	All components shall timestamp in UTC, taken from GNSS where available.	S	T	BENCH	01
SWE-009	Mission Control software shall start automatically within 60 s of power-on.	S	T	BENCH	04

5.18 Logging and data (LOG)

ID	Requirement	Pri	Ver	Stage	Trace
LOG-001	The helm shall log position, attitude, mode, commands, battery and failsafe events at 5 Hz or faster.	M	T	BENCH	01
LOG-002	Mission Control shall log: all commands, sign-offs and approvals; each mission as approved (JSON); language model requests and replies; and validator results.	M	T	SIM	01, 08
LOG-003	The logs shall be enough to replay any trip on a map, including every failsafe event.	M	D	SIM	01
LOG-004	Photos shall be stored locally and never published automatically. Photos go to cloud services only if an adult enables cloud analysis for that session.	M	I, T	SIM	07
LOG-005	Logs and photos may be backed up at home and kept for at least 12 months.	C	I	SIM	03

5.19 Environment (ENV)

ID	Requirement	Pri	Ver	Stage	Trace
ENV-001	The system shall operate in air at 0 to 35 °C and in fresh water at 2 to 25 °C.	M	A, T	LAKE	01
ENV-002	The boat shall make at least 0.5 m/s headway into a 5.4 m/s wind (top of Beaufort 3).	M	T	LAKE	01
ENV-003	The boat shall operate in waves up to 100 mm.	M	T	LAKE	01
ENV-004	The boat should keep operating in light rain. Heavy rain is outside the operating envelope.	S	T	POOL	01
ENV-005	The boat shall keep operating in sparse surface weed, recovering through FS-005 and FS-006.	S	T	LAKE	01
ENV-006	Exposed parts shall keep their function over two seasons of outdoor use.	C	I	LAKE	05

5.20 Child suitability (CHD)

Design rules for everything the 4-year-old crew member handles.

ID	Requirement	Pri	Ver	Stage	Trace
CHD-001	No child-handled part shall fit wholly inside the EN 71-1 small-parts cylinder.	M	T	BENCH	04, 06
CHD-002	Child-handled parts shall have rounded edges (radius at least 0.5 mm) and no pinch points.	M	I	BENCH	04, 06
CHD-003	The battery, electronics and propellers shall not be reachable without tools or an adult action.	M	I	BENCH	06
CHD-004	'Child tier' assembly (hull segments, pods, deck, flag) shall need no tools and be doable with small hands.	M	D	BENCH	04
CHD-005	The crew controls (GO, COME HOME, STOP) shall be usable without reading, by colour and icon alone.	M	D	BENCH	04
CHD-006	No child-accessible surface shall exceed 48 °C.	S	T	BENCH	06

5.21 Site and operations (OPS)

Rules for how and where the system is used. They are verified by inspecting the checklist and the permission.

ID	Requirement	Pri	Ver	Stage	Trace
OPS-001	The pre-launch checklist shall cover at least: site permission and conditions; no swim session, course or algae notice; wind at Beaufort 3 or less, blowing towards the launch bank where possible; battery charged; box sealed and latched; props and guards; fence and exclusions reviewed on the map; recovery kit present; beacon working.	M	I	POOL	06, 07, 01
OPS-002	Written permission from Cambridge Sport Lakes Trust shall be obtained before any operation at Milton Country Park, and its conditions followed.	M	I	LAKE	07
OPS-003	There shall be no operation during swim sessions or watersports courses, or while a blue-green algae warning is posted.	M	I	LAKE	06, 07
OPS-004	Exclusion zones shall keep the boat at least 20 m from occupied angling swims and other water users. If anyone comes within 20 m of the boat, the operator shall command RTL or HOLD.	M	I	LAKE	06, 07
OPS-005	Islands, reed beds and known nesting areas shall be exclusion zones. No mission shall aim to approach or follow wildlife.	M	I	LAKE	06
OPS-006	At Milton, the Mk1 inclusion fence shall stay within 100 m of the launch point until at least 5 successful lake missions have been logged.	S	I	LAKE	01, 07
OPS-007	An adult operator shall be present and in control throughout. The child shall never be unsupervised near the water.	M	I	POOL	06
OPS-008	The recovery kit shall include a telescopic pole of at least 3 m with a landing net, and a casting rod with a weighted line. Nobody shall enter the water to recover the boat.	M	I	LAKE	01, 06
OPS-009	Trials shall progress: simulation, then bench, then paddling pool or small water, then lake. Each failsafe shall be demonstrated before the first lake trial.	M	I	LAKE	01
OPS-010	Batteries shall be charged under supervision in a fire-safe bag, and stored at storage voltage.	M	I	BENCH	06
OPS-011	A spares kit shall be taken to every session: spare propellers, a spare fuse, cable ties, a microfibre lens cloth and a towel.	S	I	LAKE	01
OPS-012	Before the first lake trial, the operator shall rehearse every contingency scenario in section 3.5 in simulation.	M	D	SIM	01, 06

5.22 Cost and constraints (CON)

ID	Requirement	Pri	Ver	Stage	Trace
CON-001	The Mk1 bill of materials shall not exceed £120, excluding items already owned (Pi 5, printer, tools, charger). The target is £100.	M	A	BENCH	05
CON-002	The owned Raspberry Pi 5 and Ultimaker printer shall be used.	M	I	BENCH	05
CON-003	Purchased components shall be available from UK retailers or common online marketplaces.	S	I	BENCH	05
CON-004	Purchased electronics shall carry CE or UKCA marking where such products are normally marked.	S	I	BENCH	07

6. Verification

6.1 Approach

Each requirement is verified at the earliest stage that can prove it. It is then re-checked at later stages when the hardware or software it depends on changes. A stage is passed only when all its Must requirements are verified. **No lake trial happens until every SIM, BENCH and POOL Must requirement has passed** (OPS-009).

Stage	Where	What it proves
SIM	Laptop / Pi 5	Logic, modes, failsafes, validator, Mission Control (SWE-004/005)
BENCH	Workshop	Hardware, electrical, ingress, child-suitability, timings
POOL	Paddling pool, then small private water	Buoyancy, stability, driving, stuck detection, link, photos
LAKE	Milton Country Park (with permission)	Performance, range, endurance, environment, end-to-end missions

6.2 Verification summary

Stage	Test	Demo	Inspect	Analysis	Sim	Reqs
SIM	43	16	12	3	23	76
BENCH	30	5	27	2	-	62
POOL	13	6	3	1	1	22
LAKE	13	2	9	1	-	24

Counts of verification methods by first stage. A requirement can have more than one method.

6.3 Requirements by first verification stage

SIM (76): MOD-001, MOD-004, MOD-005, MOD-006, MOD-007, MOD-008, PRE-002, PRE-005, NAV-004, NAV-007, NAV-008, FEN-001, FEN-002, FEN-003, FEN-004, FEN-005, FEN-006, FEN-007, FS-001, FS-002, FS-003, FS-004, FS-007, FS-011, FS-012, FS-013, SAF-002, SAF-003, SAF-004, SAF-006, SAF-007, MIS-001, MIS-002, MIS-003, MIS-004, MIS-005, MIS-006, NLI-001, NLI-002, NLI-003, NLI-004, NLI-005, NLI-006, NLI-008, VAL-001, VAL-002, VAL-003, VAL-004, VAL-005, VAL-006, VAL-007, VAL-008, VAL-009, VAL-010, DET-004, MC-003, MC-005, MC-007, MC-008, MC-009, MC-010, MC-011, MC-015, MC-014, SWE-001, SWE-002, SWE-003, SWE-004, SWE-005, SWE-006, SWE-007, LOG-002, LOG-003, LOG-004, LOG-005, OPS-012

BENCH (62): MOD-002, MOD-003, PRE-001, PRE-003, PRE-004, PRE-007, PRE-008, FS-008, FS-009, FS-010, SAF-001, SAF-005, CAM-001, CAM-002, CAM-003, CAM-004, MC-001, MC-002, MC-004, MC-013, COM-001, COM-006, REC-002, REC-003, REC-006, REC-007, MEC-001, MEC-002, MEC-005, MEC-006, MEC-007, MEC-008, MEC-009, MEC-010, MEC-012, MEC-013, MEC-014, MEC-015, PWR-001, PWR-002, PWR-003, PWR-004, PWR-005, PWR-006, PWR-007, PWR-008, PWR-010, PWR-011, SWE-008, SWE-009, LOG-001, CHD-001, CHD-002, CHD-003, CHD-004, CHD-005, CHD-006, OPS-010, CON-001, CON-002, CON-003, CON-004

POOL (22): PRE-006, NAV-001, NAV-003, FS-005, FS-006, CAM-005, CAM-006, CAM-007, CAM-008, DET-003, MC-012, COM-003, COM-004, COM-005, COM-007, REC-001, REC-005, MEC-003, MEC-004, ENV-004, OPS-001, OPS-007

LAKE (24): NAV-002, NAV-005, NAV-006, NLI-007, DET-001, DET-002, MC-006, COM-002, REC-004, MEC-011, PWR-009, ENV-001, ENV-002, ENV-003, ENV-005, ENV-006, OPS-002, OPS-003, OPS-004, OPS-005, OPS-006, OPS-008, OPS-009, OPS-011

7. Traceability

Every stakeholder need is covered by at least one requirement, and every requirement traces to at least one need. The build script checks both automatically. The matrix counts requirements per area and need.

Area	01	02	03	04	05	06	07	08
MOD	5			1		6		
PRE	7	1				1	2	
NAV	5	2	2			3	1	
FEN	7					3	1	
FS	12					4		
SAF	7	1				1		3
MIS	2	4	1	1				1
NLI	1	7		2		1		1
VAL	9	2	1			2		1
CAM	1		7	1				
DET			3	1		1		
MC	8	2	1	3	1	5	1	
COM	6	1				1	1	
REC	7						1	
MEC	7			5	3	1		
PWR	5		1		1	5		
SWE	3			1				6
LOG	3		1				1	1
ENV	5				1			
CHD				4		4		
OPS	6					8	5	
CON					3		1	
Total	106	20	17	19	9	46	14	13

01 Never lose the boat · **02** Plain-English instructions · **03** Photos of ducks · **04** A build a 4-year-old can join · **05** Affordable · **06** Safe for people and wildlife · **07** Welcome at the lake · **08** Python-first and extensible

7.1 Requirements per need

STK-01 Never lose the boat (106): MOD-001, MOD-004, MOD-005, MOD-006, MOD-007, PRE-001, PRE-002, PRE-003, PRE-004, PRE-005, PRE-007, PRE-008, NAV-001, NAV-005, NAV-006, NAV-007, NAV-008, FEN-001, FEN-002, FEN-003, FEN-004, FEN-005, FEN-006, FEN-007, FS-001, FS-002, FS-003, FS-004, FS-005, FS-006, FS-007, FS-008, FS-010, FS-011, FS-012, FS-013, SAF-001, SAF-002, SAF-003, SAF-004, SAF-005, SAF-006, SAF-007, MIS-002, MIS-003, NLI-004, VAL-001, VAL-002, VAL-003, VAL-004, VAL-005, VAL-006, VAL-008, VAL-009, VAL-010, CAM-004, MC-003, MC-005, MC-006, MC-009, MC-011, MC-012, MC-013, MC-014, COM-002, COM-003, COM-004, COM-005, COM-006, COM-007, REC-001, REC-002, REC-003, REC-004, REC-005, REC-006, REC-007, MEC-002, MEC-004, MEC-006, MEC-011, MEC-012, MEC-013, MEC-015, PWR-005, PWR-006, PWR-009, PWR-010, PWR-011, SWE-004, SWE-005, SWE-008, LOG-001, LOG-002, LOG-003, ENV-001, ENV-002, ENV-003, ENV-004, ENV-005, OPS-001, OPS-006, OPS-008, OPS-009, OPS-011, OPS-012

STK-02 Plain-English instructions (20): PRE-005, NAV-004, NAV-005, SAF-004, MIS-001, MIS-004, MIS-005, MIS-006, NLI-001, NLI-002, NLI-003, NLI-004, NLI-005, NLI-006, NLI-007, VAL-001, VAL-007, MC-005, MC-011, COM-005

STK-03 Photos of ducks (17): NAV-002, NAV-006, MIS-001, VAL-005, CAM-001, CAM-002, CAM-003, CAM-004, CAM-005, CAM-006, CAM-007, DET-001, DET-002, DET-003, MC-015, PWR-009, LOG-005

STK-04 A build a 4-year-old can join (19): MOD-007, MIS-005, NLI-002, NLI-005, CAM-008, DET-002, MC-002, MC-007, MC-015, MEC-001, MEC-003, MEC-007, MEC-008, MEC-009, SWE-009, CHD-001, CHD-002, CHD-004, CHD-005

STK-05 Affordable (9): MC-001, MEC-001, MEC-005, MEC-014, PWR-001, ENV-006, CON-001, CON-002, CON-003

STK-06 Safe for people and wildlife (46): MOD-001, MOD-002, MOD-003, MOD-004, MOD-006, MOD-008, PRE-006, NAV-002, NAV-003, NAV-007, FEN-001, FEN-004, FEN-006, FS-008, FS-009, FS-011, FS-013, SAF-006, NLI-006, VAL-002, VAL-008, DET-004, MC-002, MC-003, MC-004, MC-008, MC-010, COM-006, MEC-010, PWR-002, PWR-003, PWR-004, PWR-007, PWR-008, CHD-001, CHD-002, CHD-003, CHD-006, OPS-001, OPS-003, OPS-004, OPS-005, OPS-007, OPS-008, OPS-010, OPS-012

STK-07 Welcome at the lake (14): PRE-002, PRE-006, NAV-003, FEN-002, MC-010, COM-001, REC-006, LOG-004, OPS-001, OPS-002, OPS-003, OPS-004, OPS-006, CON-004

STK-08 Python-first and extensible (13): SAF-001, SAF-003, SAF-005, MIS-004, NLI-008, VAL-006, SWE-001, SWE-002, SWE-003, SWE-004, SWE-006, SWE-007, LOG-002

8. Open items (TBD)

ID	Open item	Affects	Resolved by
TBD-01	Software route: A (Python on ArduPilot) or B (pure-Python helm + MicroPython guardian).	SAF-005, SWE-003, MC-014	Parent decision
TBD-02	Which Milton lake, the launch point and home-bay coordinates.	OPS-006, FEN-002	Site visit
TBD-03	Cambridge Sport Lakes Trust permission and conditions.	OPS-002, OPS-003	Email the Trust
TBD-04	Wi-Fi setup between boat, Pi 5 and phone hotspot.	COM-001, COM-005	Architecture
TBD-05	Speech-to-text engine: on-device (Pi 5) or cloud.	NLI-002	Architecture
TBD-06	Display device: attached screen, tablet or phone.	MC-005, MC-006	Architecture
TBD-07	Battery: 3S Li-ion pack or 3S LiPo.	PWR-001	Architecture
TBD-08	Failsafe thresholds and timings: tune in simulation and pool trials.	FS-001 to FS-006	Trials

After review: I'll fold in your comments as Issue B. Once TBD-01 (software route) is decided, the architecture design allocates every requirement to a component (boat hardware, helm, mission computer, Mission Control, procedures) and adds interface specifications.

Appendix A. Glossary

Term	Meaning
Helm	The part of the system that steers and enforces the fence: ArduPilot on a flight controller (Route A) or a Python autopilot plus guardian (Route B).
Mission computer	The on-boat companion computer: camera, photos, link bridge, higher-level behaviour.
Mission Control	The Raspberry Pi 5 bank station with GO, COME HOME and STOP buttons.
Guardian	Route B only: an independent MicroPython microcontroller that gates motor commands.
Geofence	An inclusion polygon the boat must stay inside, plus exclusion zones it must stay out of.
RTL	Return To Launch: navigate home and hold.
HOLD	Stay put: station-keeping, or motors off after STOP or some failsafes.
SoC	State of charge of the battery, in %.
HDOP	Horizontal dilution of precision, a GNSS quality measure. Lower is better.
MAVLink	The message protocol spoken by ArduPilot and ground stations.
SITL	Software In The Loop: ArduPilot running as a simulator.
FMEA	Failure Modes and Effects Analysis.
Operator	The responsible adult.
Crew	The 4-year-old co-pilot.

Appendix B. Requirement statistics

Area	Title	Must	Should	Could	Total
MOD	Operating modes	6	1	1	8
PRE	Pre-arm checks	8	-	-	8
NAV	Navigation and control	5	3	-	8
FEN	Geofence	6	1	-	7
FS	Failsafes	11	2	-	13

Area	Title	Must	Should	Could	Total
SAF	Safety architecture	6	1	-	7
MIS	Missions	5	1	-	6
NLI	Natural-language instructions	6	2	-	8
VAL	Mission validation	10	-	-	10
CAM	Camera and photos	4	3	1	8
DET	Duck finding and captain's log	1	2	1	4
MC	Mission Control (bank station)	8	7	-	15
COM	Communications	5	1	1	7
REC	Recoverability and locating	5	1	1	7
MEC	Mechanical	11	4	-	15
PWR	Electrical and power	10	1	-	11
SWE	Software	6	3	-	9
LOG	Logging and data	4	-	1	5
ENV	Environment	3	2	1	6
CHD	Child suitability	5	1	-	6
OPS	Site and operations	10	2	-	12
CON	Cost and constraints	2	2	-	4
	Total	137	40	7	184